

Администрирование сетевых подсистем

Лабораторная работа №15

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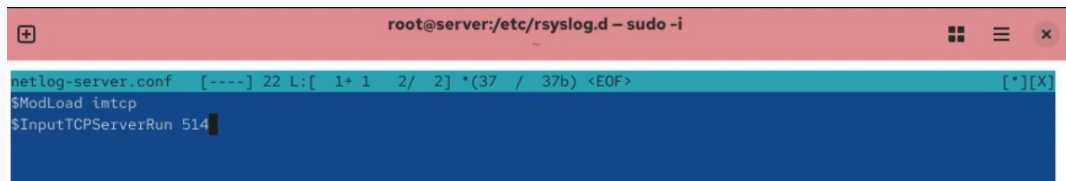
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Цели и задачи работы

Получение практических навыков по настройке системы сетевого журналирования, работе с rsyslog и анализу логов.

Выполнение лабораторной работы

Настройка сервера сетевого журнала



The image shows a terminal window with a pink title bar. The title bar contains a plus icon on the left, the text "root@server:/etc/rsyslog.d - sudo -i" in the center, and window control icons (four squares, a hamburger menu, and a close 'x') on the right. The terminal content is as follows:

```
netlog-server.conf [----] 22 L:[ 1+ 1 2/ 2] *(37 / 37b) <EOF> [*][X]  
$ModLoad imtcp  
$InputTCPServerRun 514
```

Рис. 1: Файл конфигурации netlog-server.conf

Прослушиваемые TCP-порты rsyslog

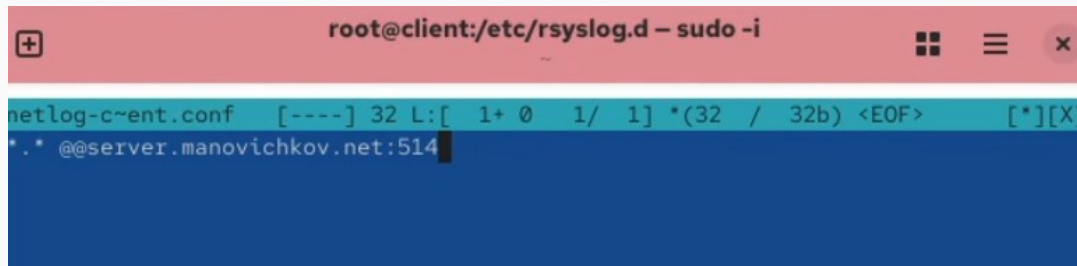
```
root@server:/etc/rsyslog.d -- sudo -i

rsyslogd 11432 root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11434 in:imjour root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11434 in:imjour root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11435 in:imtcp root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11435 in:imtcp root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11436 in:imtcp root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11436 in:imtcp root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11437 in:imtcp root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11437 in:imtcp root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11438 in:imtcp root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11438 in:imtcp root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11439 in:imtcp root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11439 in:imtcp root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11440 rs:main root 4u IPv4 65939 0t0 TCP *:shell (LISTEN)
rsyslogd 11432 11440 rs:main root 5u IPv6 65940 0t0 TCP *:shell (LISTEN)

[root@server.manovichkov.net rsyslog.d]# firewall-cmd --add-port=514/tcp
success
[root@server.manovichkov.net rsyslog.d]# firewall-cmd --add-port=514/tcp --permanent
success
[root@server.manovichkov.net rsyslog.d]#
```

Рис. 2: Прослушиваемые порты rsyslog

Настройка клиента сетевого журнала



The image shows a terminal window with a red title bar. The title bar contains a plus icon on the left, the text "root@client:/etc/rsyslog.d – sudo -i" in the center, and window control icons (four squares, three horizontal lines, and a close 'x') on the right. The terminal content has a light blue background. The first line of text is "netlog-c~ent.conf [----] 32 L:[1+ 0 1/ 1] *(32 / 32b) <EOF> [*][X]". The second line shows the configuration line "*. * @@server.manovichkov.net:514" followed by a black cursor block.

```
root@client:/etc/rsyslog.d – sudo -i  
netlog-c~ent.conf [----] 32 L:[ 1+ 0 1/ 1] *(32 / 32b) <EOF> [*][X]  
*. * @@server.manovichkov.net:514
```

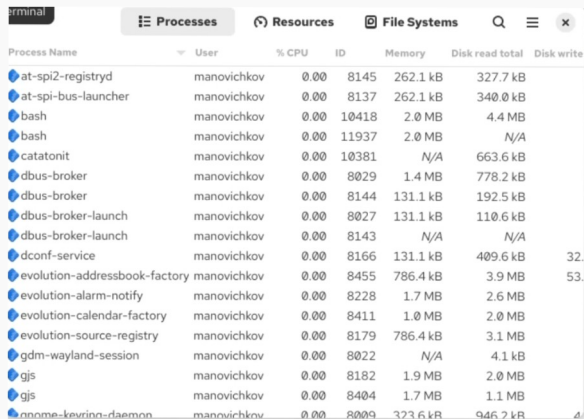
Рис. 3: Файл netlog-client.conf

Просмотр поступающих журналов

```
root@server:/etc/rsyslog.d - sudo -i

Nov  5 10:31:51 client kernel: traps: VBoxClient[30480] trap int3 ip:41ddb sp:7f00a01b4cd0 error:0 in VBoxClient[1dd1b,400000+bb000]
Nov  5 10:31:51 client systemd-coredump[30481]: Process 30477 (VBoxClient) of user 1001 terminated abnormally with signal 5/TRAP, processing...
Nov  5 10:31:51 client systemd[1]: Started systemd-coredump@80-30481-0.service - Process Core Dump (PID 30481/UID 0).
Nov  5 10:31:51 client systemd-coredump[30482]: Process 30477 (VBoxClient) of user 1001 dumped core.#012#012Module 1ibXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 30480:#012#0 0x000000000041dd1b n/a (n/a + 0x0)#012#1 0x000000000041dc94 n/a (n/a + 0x0)#012#2 0x000000000045041c n/a (n/a + 0x0)#012#3 0x00000000004355d0 n/a (n/a + 0x0)#012#4 0x00007f00ae8b7b68 start_thread (libc.so.6 + 0x94b68)#012#5 0x00007f00ae9286bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 30477:#012#0 0x00007f00ae9264bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004344e2 n/a (n/a + 0x0)#012#2 0x0000000000450066 n/a (n/a + 0x0)#012#3 0x000000000000405123 n/a (n/a + 0x0)#012#4 0x00007f00ae84d30e __libc_start_call_main (libc.so.6 + 0x2a30e)#012#5 0x00007f00ae84d3c9 __libc_start_main@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6 0x00000000004044aa n/a (n/a + 0x0)#012ELF object binary architecture: AMD x86-64
Nov  5 10:31:51 client systemd[1]: systemd-coredump@80-30481-0.service: Deactivated successfully.
Nov  5 10:31:53 server kernel: traps: VBoxClient[11890] trap int3 ip:41ddb sp:7f24624fbcd0 error:0 in VBoxClient[1dd1b,400000+bb000]
Nov  5 10:31:53 server systemd-coredump[11891]: Process 11887 (VBoxClient) of user 1001 terminated abnormally with signal 5/TRAP, processing...
Nov  5 10:31:53 server systemd[1]: Started systemd-coredump@165-11891-0.service - Process Core Dump (PID 11891/UID 0).
Nov  5 10:31:53 server systemd-coredump[11892]: Process 11887 (VBoxClient) of user 1001 dumped core.#012#012Module 1ibXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 11890:#012#0 0x000000000041dd1b n/a (n/a + 0x0)#012#1 0x000000000041dc94 n/a (n/a + 0x0)#012#2 0x000000000045041c n/a (n/a + 0x0)#012#3 0x00000000004355d0 n/a (n/a + 0x0)#012#4 0x00007f2470baf68 start_thread (libc.so.6 + 0x94b68)#012#5 0x00007f2470c286bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 11887:#012#0 0x00007f2470c1e4bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004344e2 n/a (n/a + 0x0)#012#2 0x0000000000450066 n/a (n/a + 0x0)#012#3 0x000000000000405123 n/a (n/a + 0x0)#012#4 0x00007f2470b4530e __libc_start_call_main (libc.so.6 + 0x2a30e)#012#5 0x00007f2470b453c9 __libc_start_main@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6 0x00000000004044aa n/a (n/a + 0x0)#012ELF object binary architecture: AMD x86-64
Nov  5 10:31:53 server systemd[1]: systemd-coredump@165-11891-0.service: Deactivated successfully.
```

Рис. 4: Просмотр сообщений журнала



The screenshot shows the 'Processes' tab in the GNOME System Monitor application. The window has a title bar with 'terminal' and a close button. The interface includes tabs for 'Processes', 'Resources', and 'File Systems', along with search and menu icons. The main area displays a table of running processes.

Process Name	User	% CPU	ID	Memory	Disk read total	Disk write
at-spi2-registryd	manovichkov	0.00	8145	262.1 kB	327.7 kB	
at-spi-bus-launcher	manovichkov	0.00	8137	262.1 kB	340.0 kB	
bash	manovichkov	0.00	10418	2.0 MB	4.4 MB	
bash	manovichkov	0.00	11937	2.0 MB	N/A	
catatonit	manovichkov	0.00	10381	N/A	663.6 kB	
dbus-broker	manovichkov	0.00	8029	1.4 MB	778.2 kB	
dbus-broker	manovichkov	0.00	8144	131.1 kB	192.5 kB	
dbus-broker-launch	manovichkov	0.00	8027	131.1 kB	110.6 kB	
dbus-broker-launch	manovichkov	0.00	8143	N/A	N/A	
dconf-service	manovichkov	0.00	8166	131.1 kB	409.6 kB	32.0 kB
evolution-addressbook-factory	manovichkov	0.00	8455	786.4 kB	3.9 MB	53.0 kB
evolution-alarm-notify	manovichkov	0.00	8228	1.7 MB	2.6 MB	
evolution-calendar-factory	manovichkov	0.00	8411	1.0 MB	2.0 MB	
evolution-source-registry	manovichkov	0.00	8179	786.4 kB	3.1 MB	
gdm-wayland-session	manovichkov	0.00	8022	N/A	4.1 kB	
gjs	manovichkov	0.00	8182	1.9 MB	2.0 MB	
gjs	manovichkov	0.00	8404	1.7 MB	1.1 MB	
gnome-keyring-daemon	manovichkov	0.00	8009	323.6 kB	946.2 kB	4.0 kB

Рис. 5: Процессы в gnome-system-monitor

```
[root@server.manovichkov.net rsyslog.d]#  
[root@server.manovichkov.net rsyslog.d]# cd /vagrant/provision/server/  
[root@server.manovichkov.net server]# mkdir -p /vagrant/provision/server/netlog/etc/rsyslog.d  
[root@server.manovichkov.net server]# cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/provision/server/netlog/etc/rsyslog.d  
[root@server.manovichkov.net server]# touch netlog.sh
```

Рис. 6: Структура каталогов сервера

Подготовка окружения Vagrant (client)

```
[root@client.manovichkov.net rsyslog.d]# systemctl restart rsyslog.service
[root@client.manovichkov.net rsyslog.d]#
[root@client.manovichkov.net rsyslog.d]# cd /vagrant/provision/client/
[root@client.manovichkov.net client]# mkdir -p /vagrant/provision/client/netlog/
etc/rsyslog.d
[root@client.manovichkov.net client]# cp -R /etc/rsyslog.d/vagrant/provision/cli
ent/netlog/etc/rsyslog.d
rsyslog.d/
[root@client.manovichkov.net client]# cp -R /etc/rsyslog.d/netlog-client.conf /v
agrant/provision/client/netlog/etc/rsyslog.d
[root@client.manovichkov.net client]# touch netlog.sh
[root@client.manovichkov.net client]#
```

Рис. 7: Структура каталогов клиента

Результаты работы

В ходе лабораторной работы была выполнена комплексная настройка системы сетевого журналирования на основе rsyslog.

Сервер был подготовлен для приёма удалённых сообщений, открыт порт 514/TCP, настроен firewall.

Клиент был сконфигурирован для пересылки всех журналов на центральный сервер.

Была проверена корректность передачи логов и выполнен анализ поступающих сообщений.

Подготовлены каталоги и скрипты для автоматизации через Vagrant.